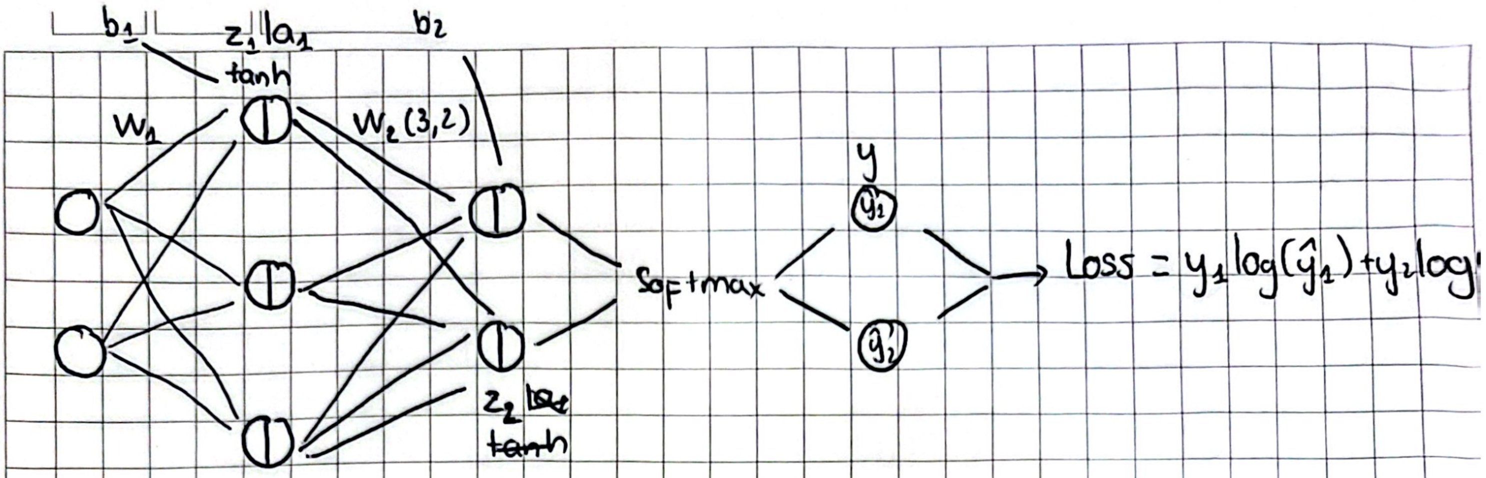

$$\cancel{\delta_3 = \frac{\partial L}{\partial y} = \hat{y} - y \ (n\times 2)} \qquad \delta_3 = \frac{\partial y}{\partial y}\cdot\frac{\partial L}{\partial z_2} = \hat{y} - y \ (n\times 2)$$

$$\underset{(3\times 2)}{\frac{\partial L}{\partial W_2}} = \frac{\partial L}{\partial y}\cdot\frac{\partial y}{\partial W_2} = a_1^T \cdot \delta_3 \qquad (a_1 : n\times 3)$$

$$\underset{(1\times 2)}{\frac{\partial L}{\partial b_2}} = \underset{(n\times 2)}{\frac{\partial L}{\partial y}}\cdot\underset{(n\times n) = I_n}{\frac{\partial y}{\partial b_2}} = \frac{\partial L}{\partial y} = \frac{\partial L}{\partial y} = \hat{y} - y = \underset{(n\times 2)}{\delta_3} \quad \to \text{sum, axis} = 0$$

$$\delta_2 = \underset{(n\times 3)}{\frac{\partial L}{\partial z_1}} = \frac{\partial L}{\partial y}\cdot\frac{\partial y}{\partial z_2}\cdot\frac{\partial z_2}{\partial a_1}\cdot\frac{\partial a_1}{\partial z_1} = \underset{(n\times 3)}{(1 - \tanh^2 z_1)} \otimes \underbrace{\underset{(n\times 2)}{\delta_3}\ \underset{(2\times 3)}{W_2^T}}_{n\times 3}$$

$$\underset{(2\times 3)}{\frac{\partial L}{\partial W_1}} = \underset{(n\times 2)^T}{x^T}\cdot \delta_2 \quad ; \quad \frac{\partial L}{\partial b_1} = \delta_2$$